

TASK 1: WEB SCRAPING USING PYTHON

INTRODUCTION

Web Scraping was performed using Python to extract useful information from websites and convert unstructured web data into a structured dataset.

The main objective of this project is to automatically collect required information from a website, process the extracted data, and store it in an organized format for further analysis.

Tools:

- Python
- Visual Studio Code
- Requests Library
- BeautifulSoup Library
- Pandas
- Open PyXL

DATASET DESCRIPTION

The dataset was created by extracting information from a website using Python web scraping techniques.

The scraped dataset contains book-related information such as:

- Book Title
- Price
- Rating

The extracted data was converted into an Excel file for further use and analysis.

STEPS PERFORMED

1. Imported required libraries

Python libraries such as Requests, BeautifulSoup, and Pandas were imported to perform web scraping and data processing.

2. Connected to the website

The Requests library was used to send a request to the website and collect webpage content.

3. Extracted webpage information

Beautiful Soup was used to analyze the HTML structure of the webpage and extract required details.

4. Collected required data

Important information such as book names, prices, and ratings were extracted from the website.

5. Processed extracted data

The collected information was converted into a structured table using Pandas Data Frame.

6. Exported dataset

The extracted data was saved into an Excel file for further analysis.

7. Verified scraped data

The generated dataset was checked to ensure that the required information was collected successfully.

OBSERVATIONS

- Website data was successfully extracted using Python.
- Unstructured web information was converted into a structured dataset.
- The scraped dataset contains useful book details.

- The generated Excel file can be used for further analysis and visualization.

CONCLUSION

Web Scraping was successfully performed using Python libraries.

The project helped in understanding how data can be automatically collected from websites and transformed into a structured format.

This process reduces manual data collection and provides datasets that can be used for further data analytics tasks.

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